

A HETEROGENEOUS ROUTING PROTOCOL BASED ON A NEW STABLE CLUSTERING SCHEME*

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ABSTRACT

A Mobile Ad Hoc Network (MANET) is usually assumed to be homogeneous, where each mobile node shares the same radio capacity. However, a homogeneous ad hoc network suffers from poor scalability. Recent research has demonstrated its performance bottleneck both theoretically and through simulation experiments. Building a physically hierarchical ad hoc network is a very promising way to achieve scalability. Routing is critical to operate such a hierarchical structure efficiently. Previous research has been solely focusing on extending popular routing schemes developed for the homogeneous ad hoc network. In this paper, we propose a new heterogeneous routing protocol specifically designed for the hierarchical ad hoc network. We also present a new active clustering scheme to help build a stable hierarchical structure, which is fundamental to heterogeneous routing protocols. Simulation results using GloMoSim show that our routing and clustering scheme gives good performance. The results also provide some insight into new possible designs of routing protocols and mobile network architectures for MANETs when large scale is desired.

INTRODUCTION

The mobile ad hoc network (MANET) [1] is getting more and more attractive because of its independence of a fixed infrastructure and its instant deployment and easy reconfiguration capabilities. A considerable body of research has been done in this field, especially on ad hoc routing schemes. Most routing protocols in MANETs are designed for a homogeneous wireless network, where all nodes share the same radio capacity. It has been proven that current routing protocols work well in small size networks (e.g. fewer than 100 nodes) [10][11]. However, recent study also reported that the “flat” ad hoc network structure has limited capacity [6]. As the size of network increases, the overall network capacity decreases substantially, especially when current routing schemes are applied [8]. Current routing schemes usually need to propagate certain routing packets throughout the whole

network. Data packets are usually routed through “long hop” paths. The “long hop” paths in large-scale network are prone to break. All these features prevent the “flat” ad hoc network structure from scaling to large scale. Its performance will degrade quickly along with the increase of network size.

However, the large-scale ad hoc networks are more and more desired in many application scenarios, especially the mobile military networks. An emerging promising solution to remove the performance bottleneck of an ad hoc network in large scale is to build hierarchical network architecture. A general picture of a two level hierarchical ad hoc network is demonstrated in Figure 1. We refer it as an ad hoc network with mobile backbones (MBN). Among the mobile nodes, some nodes, named backbone nodes (BNs), have an additional powerful radio to establish wireless links among themselves. Thus, forming a higher-level network called the backbone network. Since the backbone nodes are also moving and join or leave the backbone network dynamically, the backbone network is exactly an ad hoc network in a different radio level.

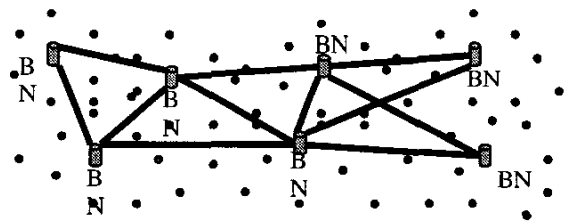


Figure 1. A picture of a two level hierarchical ad hoc network

Routing is critical to effectively and efficiently operate such a hierarchical ad hoc network. So far, several routing schemes targeting the physically hierarchical structure have been proposed [7][9]. All of them are simply the extension of popular routing schemes originally designed for “flat” ad hoc networks. A homogeneous routing protocol is applied to support routing. In this paper, we present a new idea of designing routing schemes for a hierarchical ad hoc network. We group mobile nodes into

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local subnets and apply two independent routing protocols in the backbone network and in local subnets. Our scheme is quite similar to current Internet routing architecture, where different routing protocols are used within autonomous systems (Intra-AS routing, e.g. OSPF) and across autonomous systems (Inter-AS routing, e.g. BGP). With a heterogeneous routing scheme, we are able to control the propagation scope of routing packets and achieve good scalability.

Unlike the Internet, where the ASs are pre-decided and gateways are configured manually, in a MANET, we have to dynamically create and maintain the local subnets (a group of mobile nodes near to each other) because of node mobility. Using clustering schemes would be a natural choice since clustering has already been widely used to form logically hierarchical networks [4]. It is suitable for partitioning the large-scale network into small groups. However, a big drawback of current clustering schemes is the instability of clusters, as indicated in many papers such as [4]. Traditional clustering schemes work effectively only in networks with very low mobility, such as the sensor network. Instability of clusters would make the hierarchy too dynamic to be operated successfully. Frequent change of cluster head and cluster members will waste most routing information. In this paper, we will present a new fully distributed clustering scheme to achieve good stability.

Rest of this paper is organized as follow. In section 2, we briefly explain why dynamic clustering is required to build the physically hierarchical ad hoc network, and introduce our new stable clustering scheme. In section 3, we compare our clustering scheme with other popular ones regarding stability. Section 4, we present our idea of heterogeneous routing and describe one heterogeneous routing protocol in detail. In section 5, our routing protocol is evaluated in a large-scale ad hoc network. Related work is given in section 6 and we conclude our paper in section 7.

CLUSTERING AND BACKBONE FORMATION

From Figure 1 we can see that the simplest way to build the hierarchical ad hoc network is to pre-assign backbone nodes and scatter them around the terrain. Each backbone node broadcast a beacon periodically. The ordinary nodes listen to beacons and select the nearest backbone node to form different subnets around each backbone node. However, such a scheme has two main problems. First, the backbone nodes are also moving, thus after sometime, some backbone nodes may move together (interfering to each other) while some areas may lack for backbone nodes. This certainly is not a good scenario. The second problem is fault tolerance. Backbone nodes may fail of

operation or even be destroyed (considering the proposed application environment of the MANET). New backbone nodes should be deployed to replace the defunct ones. Static deployment cannot fulfill these requirements. To solve these two problems, our solution is to deploy redundant backbone capable nodes. Here, backbone capable nodes mean those mobile nodes, which have the physical radio capacity to communicate with other backbone nodes and form the backbone network. Since backbone capable nodes are more than really needed, only part of them really become backbone nodes. Others are kept as redundancy. When one backbone node is destroyed or moves out an area, new backbone node will be selected from the backbone capable nodes to replace the old one. When two backbone nodes move near to each other, one of them will give up its backbone position. The way to select backbone nodes from capable nodes is called backbone election. Backbone election should be dynamically performed and the elected backbone nodes should distribute around the terrain. Clearly performing clustering among the backbone capable nodes is a good solution, as it has been widely used to form logically hierarchical ad hoc networks [4].

A. Random Competition based Clustering (RCC)

Many clustering schemes have been presented in the last several years, such as in [2][3][4]. Among them, the Lowest ID (LID) and Highest Degree (HD) algorithms are widely used due to their simplicity. The detail of the two algorithms can be found in [2]. Previous research in clustering mainly focuses on how to form clusters with a good shape such as minimum overlap of clusters, coverage etc. However, stability is also a serious problem to the real applications, especially when clustering is used to support routing. For the hierarchical structure, stability of backbone nodes and local subnets are highly preferred.

Targeting at the stability and simplicity, we designed a new Random Competition based Clustering (RCC) scheme. The main idea is that any node, which doesn't belong to any cluster, can initiate a cluster formation by broadcasting a packet to claim itself as a cluster head. The first node, which broadcast such a packet, will be elected as the cluster head by its neighbors. All its neighbor nodes, after hearing such a broadcast, give up their right to be a cluster head and become members of this cluster. Cluster heads have to periodically broadcast a cluster head claim packet to maintain their clusters. Since there is a delay from when one node broadcast its cluster head claim packet to when this packet is heard by its neighbors, several neighbor nodes may broadcast during this time period. To reduce such concurrent broadcasts, we introduce a random timer. Each node defers a random time before its cluster head claim. If it hears a cluster head

claim during this random time, it then gives up its broadcast. The idea of first claim node wins is first proposed in the passive clustering scheme in [5]. However, our scheme is active clustering and we introduce an explicit random timer to reduce conflicts. Of course, the random timer cannot completely solve the concurrent broadcast problem. When the concurrent broadcasts happen, we use the node ID to solve the conflict. The node with lower ID will become the cluster head.

Our Random Competition based Clustering (RCC) scheme is more stable than traditional clustering schemes such as LID and HD. In the LID scheme, when the cluster head hears a node with a lower ID, it will give up its cluster head role. Similarly, in the HD scheme, when a node with more neighbors appears, the cluster will also be reformed. Due to node mobility, such things may happen very frequently. In RCC, one node only gives up its cluster head position when another cluster head moves near to it. Since cluster heads are usually at least two hops away, clusters formed by RCC are much more stable.

B. Multihop Clustering Extension

Usually the clustering schemes are one hop based, that is the cluster head can reach all members in one hop. This is not suitable for building the hierarchical structure. One requirement for deploying the backbone nodes is that the approximate number of backbone nodes should be controllable. We extend clustering schemes to form K-hop clusters. Here, K-hop means that a cluster head can reach its members in at most K hops. By adjusting the parameter K, we can approximately control the number of cluster head. Bigger K means fewer cluster heads. In the hierarchical structure, it means fewer backbone nodes.

To extend our clustering scheme to be multihop, each node now has to forward the cluster head claim packet from its cluster head. A mobile node will select the nearest cluster head within its K-hop scope to be its cluster head. When there is no any cluster head within its K-hop scope, it claims itself as a cluster head after deferring random time. In multihop clustering, the probability of concurrent cluster head claim is high due to the longer time for propagating cluster head claim packet K-hop away. The random time delay plays a very important role here. A multihop cluster is illustrated in Figure 2

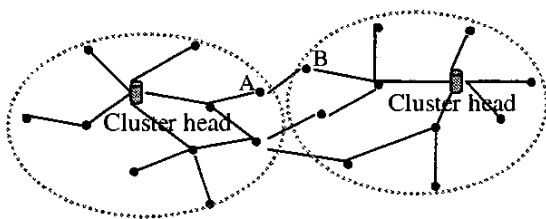


Figure 2. Multihop clusters

EVALUATION OF CLUSTER STABILITY

In this section, we use simulation to compare the stability of our Random Competition based Clustering (RCC) algorithm with the Lowest ID (LID) and Highest Degree (HD) algorithms. We use GloMoSim 2.0 [13], a packet level network simulator of ad hoc networks. Since we are targeting at large scale networks, 1000 mobile nodes are deployed. The terrain size is as large as 3200mX3200m. Each mobile node has an IEEE 802.11 wireless radio with transmission range as 200m. The DCF mode of IEEE 802.11 is used and channel bandwidth is set to 2M, following the standard. Node mobility model is random waypoint mobility [11]. In our simulation, the pause time is kept as 30 seconds and we vary the mobility speed to observe the stability of clusters. Simulation time of each run is 6 minutes.

The stability of clusters includes two parts, the stability of the cluster head and the stability of the cluster members. We define two metrics, average lifetime of a cluster head and the average membership time of a cluster member, to measure both two kinds of stability. The average lifetime of a cluster head is defined as the average time period during which one node plays the role as a cluster head continuously. The average membership time is the average time that one mobile node remains in a cluster. These two metrics fully reflect the stability of clusters. In our simulation, we only implement the basic clustering scheme without considering the “gateway” node selection as in [2]. etc.

A. On- hop Cluster Stability

Usually, clustering is performed to form one hop clusters. Thus, we first compare the stability of one-hop clusters. Here, again one hop means that the cluster head can reach all its members in one hop. The simulation results are given in Figure 3 and Figure 4.

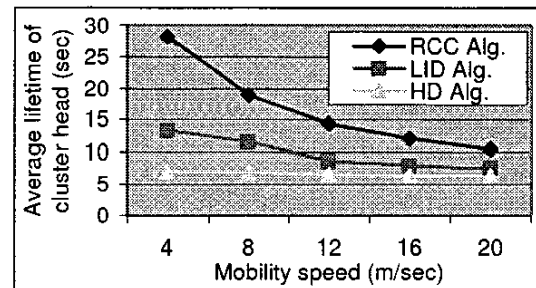


Figure 3. Average lifetime of cluster head

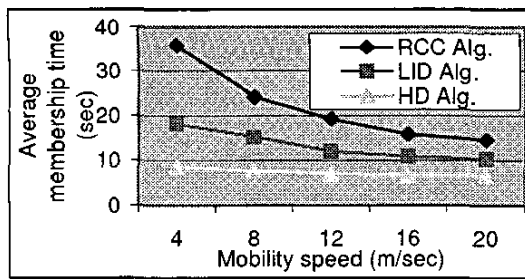


Figure 4. Average membership time

From Figure 3 and Figure 4, we can see our clustering algorithm is much stable than the lowest ID and highest degree algorithms in both low mobility and high mobility scenarios. Stability of the highest degree algorithm is the worst. This is due to the fact that the degree of one node is changing frequently under mobility.

B. Multihop Cluster Stability

By using multihop clustering, we can group mobile nodes and partition the whole network into different subnets and approximately control the subnet size. Here, the scope of multihop clustering is set to 2. We only present the results of average membership time in Figure 5.

Scheme	Low mobility (4m/sec)	High mobility (20m/sec)
RCC Alg.	57.71sec	24.05sec
LID Alg.	13.32sec	8.43sec
HD Alg.	7.41sec	4.91sec

Figure 5. Average membership time of multihop clusters

Figure 5 shows that using our scheme, the average time that one node remains in a cluster is nearly 1 minute in low mobility (4 m/sec) and 24 seconds in high mobility (20 m/sec). Since the whole simulation time is 6 minutes (360 seconds), our scheme forms very stable clusters. In contrast the average membership time of lowest ID and highest degree algorithms is decreased due the oscillations of boundary nodes. Again, our scheme outperforms Lowest ID and Highest Degree algorithms greatly.

Note, in all experiments, we assume that every mobile node has capability to be a cluster head. While clustering is applied to backbone node election, since only a small fraction (e.g. 10%) of the total mobile nodes are backbone capable, we expect clusters to be more stable.

We didn't present the results of control overhead due to page limitation. However, the low control overhead of our scheme is clear. In the lowest ID and highest degree clustering schemes, each node has to know the complete information of neighbor nodes (one hop clustering) or nodes within its K-scope (multihop clustering). In our

scheme, a node only needs to broadcast its cluster head information, which is only one entry with few fields. Boundary nodes even don't need to broadcast any packet. In one hop clustering, only the cluster head needs to broadcast control packets periodically.

HETEROGENEOUS ROUTING SCHEMES

After backbone nodes are elected via clustering, they can communicate to each other using powerful "long haul" radios and form a higher-level backbone network, which is also a dynamic ad hoc network. Previous research regarding routing in such a hierarchical structure is mainly extensions of current popular routing protocols in the flat ad hoc network [7][9]. A homogeneous routing protocol is applied to the whole network. However, such an extension cannot completely solve the scalability problem of the original routing protocol in the large-scale network. For example, the on-demand extension still needs to propagate the Route Request (RREQ) packets throughout the whole network, still facing the excessive overhead problem and wasting network bandwidth.

With the support of our clustering scheme, we can partition the large-scale network into small size subnets around backbone nodes and use the backbone node to represent each subnet. The backbone nodes can be viewed as the gateway routers connecting different subnets. To do this, we run separated routing protocols within the subnets and across subnets, which we refer as heterogeneous routing schemes. Since proactive routing and on-demand routing are the two basic categories in the MANET, generally there are four types of heterogeneous routing schemes with different combinations of them. We will use a term "Scheme1-Scheme2" to indicate one type of a heterogeneous routing scheme, for example, On-demand - Proactive means applying on-demand routing in the backbone network and proactive routing in the local subnets. All nodes in the same local subnet including the backbone node run the same routing scheme limited within the subnet. Backbone nodes, in addition, run another routing protocol in the backbone network and are in charge of routing data packets to destinations outside the local subnets. Routing packets are now limited within their local subnets or in the backbone network.

To limit routing information within the local subnet, we introduce an additional field into the routing control packets. This field stores the backbone node address of the source node, which issues this control packet. Each node only re-broadcasts routing packets, which contain the same backbone node address as itself. Generally, even different routing protocols can be used in different local subnets. Routing protocol in the backbone network behaviors exactly same as in a traditional ad hoc network, just

running on top of different network interfaces (e.g. using the powerful backbone radios).

In this paper, we only discuss one type of heterogeneous routing protocols, On-demand – Proactive, that is on-demand routing in the backbone network and proactive routing in the local subnets. In the local subnets, a proactive routing such as DSDV is applied. Thus, each node knows the exact routing information to other nodes within the same subnet. For destination nodes outside of the local subnet, data packets are simply routed to the backbone node in that subnet. The backbone node then issues RREQ packets in the backbone network to find an available route. The backbone node, which contains the destination node in the local subnet, will reply with a RREP packet. Then the data packet will be routed to the remote backbone node via backbone links. The remote backbone node finally will route the packet to its destination node within its local subnet. Note, size of the backbone network is usually limited (smaller than 50 nodes even 1000 mobile nodes are involved). On-demand routing works very well in such an environment.

PERFORMANCE EVALUATION

In this section, we evaluate our heterogeneous routing protocol, On-demand – Proactive, with a popular on-demand routing protocol, AODV [10], and its extension to the hierarchical structure. Our implementation of On-demand – Proactive protocol is AODV in the backbone network and DSDV [12] in the local subnets (AODV – DSDV). Both of them are popular ad hoc routing protocols. We also extend AODV routing protocol to utilize the physical hierarchy, which we called Hierarchical AODV (H-AODV). In H-AODV, basically both backbone nodes and ordinary nodes run the same AODV routing protocol. However, the backbone nodes will broadcast RREQ packets throughout the backbone network in addition. By recording the interface ID for each route learned, later RREP packets and data packets can also utilize the backbone links. The key point of H-AODV is that the route discovery procedure can take the “short cut” advantage of the physical hierarchy. Thus, the backbone links are usually utilized to route packets to remote destinations. We run AODV – DSDV and H-AODV in the identical environment. As a comparison, we also run the AODV routing protocol in a corresponding “flat” ad hoc network without the higher-level backbone links. Thus, no backbone election is needed. The whole network is homogeneous. All other configurations are kept exactly same.

The configuration of simulation environment is same as previous experiments in section III. Among the 1000 mobile nodes, only 10% (100) nodes are backbone

capable. Our clustering scheme is adopted to elect backbone nodes from these backbone capable nodes. Scope of clustering is 2 hops. Around 40 backbone nodes are elected during the simulation. Each backbone node has two IEEE 802.11 radios. One is same to the ordinary node with transmission range 200m. The other is more powerful with a radio range as 800m. IEEE 802.11 DCF MAC is also used in the backbone network. (Note, the two radios use different frequencies.) Backbone nodes share the same mobility model, random waypoint mobility, as ordinary mobile nodes. Traffic sources are 30 CBR (Constant Bit Rate). Source-destination pairs are randomly selected throughout the network. Data packet size is fixed as 512 bytes. Packet sending rate is 2 packet/sec. We vary the mobility speed to compare the performance.

Due to page limitation, the main metrics we selected are delivery fraction, average end-to-end delay, and average hop count of successfully delivered data packets. The delivery fraction and end-to-end data packet delay is obvious. The average hop count of successfully delivered data packets is counted using the TTL field of IP header. The average hop count is very important. Long paths (e.g. around 10 hops or more) are usually fragile and prone to break. By utilizing the backbone links, the long paths can be greatly reduced. This metric reflects that how efficiently the routing scheme takes the “short cut” advantage of the physical hierarchy. Simulation results are given from Figure 6 to Figure 8.

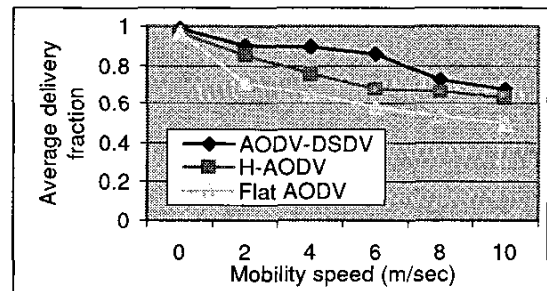


Figure 6. Average delivery fraction vs. mobility

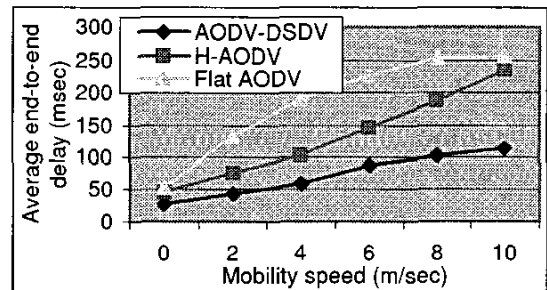


Figure 7. Average end-to-end delay vs. mobility

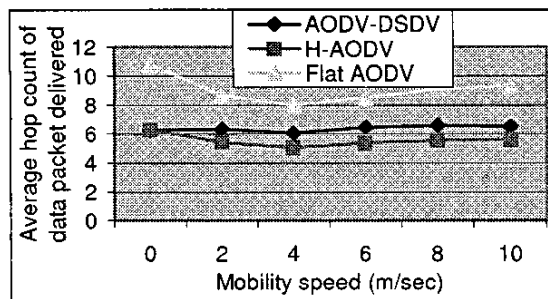


Figure 8. Average hop count of delivered packet vs. mobility

We briefly discuss results here. From Figure 6 and Figure 7 we can see that our new heterogeneous routing protocol achieve a better delivery fraction and end-to-end delay than Flat AODV and H-AODV. The long end-to-end delay of Flat AODV and H-AODV is due to the nature that AODV has to delay data packets while finding an available route. With mobility, it happens frequently. The performance improvement of H-AODV comparing to flat AODV is because H-AODV utilizes the backbone links, thus find a route quicker. The shorter routes also improve the delivery fraction.

Figure 8 proves the “short cut” of the physical hierarchy. AODV – DSDV and H-AODV have a much shorter average hop count of delivered data packets. Two things need to be addressed here. First, the AODV – DSDV scheme has a slightly longer average hop count than the H-AODV. The reason is that H-AODV usually can find the shortest paths to destinations in the network. However, the AODV – DSDV always use the backbone links to reach destinations outside of sources’ local subnets. Such a path may not always be the shortest path. However, considering the high quality of the backbone links, such a trade off is still acceptable. Second, the average hop count of AODV is actually decreased while the mobility speed is increased, although not very apparently. This is due to the reason that in the high mobility speed, data packets using the long hop paths are easy to be dropped on the way.

RELATED WORK

Several routing schemes have been proposed for the physically hierarchical ad hoc networks in the literature, such as the EHSR [9] and on-demand routing extension [7]. However, all of them apply a homogeneous routing protocol throughout the whole network. Till now, we have not seen any heterogeneous routing schemes presented. Our heterogeneous routing scheme illustrates a new way to design routing protocols for the large-scale ad hoc network. It partitions the network into subnets and limits routing control overhead in a small area. More important, our scheme has the potential to connect small ad hoc networks running different routing protocols into an

integrated large-scale network. This is valuable since in many situations, requiring each group of the large-scale network running the identical homogeneous routing scheme is very difficult and makes the network not scalable.

CONCLUSION

In this paper, we present a stable clustering scheme to dynamically build a physically hierarchical ad hoc network. A large-scale network is then divided into stable small local subnets, similar to the current Internet hierarchical structure. However, our wireless backbones are mobile backbones, thus suitable to the mobile computing environment. A new heterogeneous routing protocol is then proposed to operate such a hierarchy efficiently. Comparing with previous homogeneous schemes, our scheme achieves better scalability. Good performance is demonstrated in simulations of large-scale networks with up to 1000 mobile nodes.

REFERENCES

- [1] IETF working group: MANET, <http://www.ietf.org/html.charters/manet-charter.html>
- [2] C. R. Lin, and M. Gerla, “Adaptive Clustering for Mobile Networks,” *IEEE Journal on Selected Areas in Communications*, Vol. 15, No. 7, Sep. 1997, pp. 1265-1275.
- [3] P. Krishna, N. H. Vaidya, M. Chatterjee, and D. K. Pradhan, “A Cluster-based Approach for Routing in Dynamic Networks,” In *Proceedings of ACM SIGCOMM Computer Communication Review*, pp. 372-378, 1997.
- [4] S. Banerjee, S. Khuller, “A Clustering Scheme for Hierarchical Control in Multi-hop Wireless Networks,” *IEEE Infocom 2001*, Anchorage, Alaska, April 2001.
- [5] M. Gerla, T.J. Kwon and G. Pei, “On Demand Routing in Large Ad Hoc Wireless Networks with Passive Clustering,” *Proceedings of IEEE WCNC 2000*, Chicago, IL, Sep. 2000.
- [6] P. Gupta, and P. R. Kumar, “The Capacity of Wireless Networks,” In *IEEE Transactions on Information Theory*, vol. IT-46, no. 2, pp. 388-404, March 2000.
- [7] Y. Ko, and N. H. Vaidya, “A Routing Protocol for Physically Hierarchical Ad Hoc Networks”, Texas A&M University, Technical Report 97-010, Sep. 1997.
- [8] S. R. Das, C. E. Perkins, and E. M. Royer, “Performance Comparison of Two On-demand Routing Protocols,” In *Proceedings of IEEE INFOCOM 2000*, Tel Aviv, Israel, Mar. 2000.
- [9] D. L. Gu, G. Pei, M. Gerla, and X. Hong, “Integrated Hierarchical Routing for Heterogeneous Multi-hop Networks,” In *Proceeding of IEEE MILCOM 2000*, Los Angeles, CA, Oct. 2000.
- [10] C. E. Perkins, and E. M. Royer, “Ad-Hoc On-Demand Distance Vector Routing,” In *Proceedings of IEEE WMCSA 1999*, New Orleans, LA, Feb. 1999.
- [11] D.B. Johnson and D. A. Maltz, “Dynamic Source Routing in Ad Hoc Wireless Networks,” In *Mobile Computing*, edited by T. Imielinski and H. Korth, Chapter 5, 1996, pp.153-181.
- [12] C. Perkins, and P. Bhagwat, “Highly Dynamic Destination-Sequenced Distance-Vector Routing (DSDV) for Mobile Computers. In *Proceeding of the ACM SIGCOMM*, October 1994.
- [13] M. Takai, L. Bajaj, R. Ahuja, R. Bagrodia, and M. Gerla, “GloMoSim: A Scalable Network Simulation Environment,” Technical report 990027, UCLA, Computer Science Department, 1999.